A TRACE AND MEASURE CRITERION FOR THE RIEMANN HYPOTHESIS FROM THE ZETA SPECTRAL TRIPLE

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Abstract. The zeta spectral triple attaches to ζ two spectral measures: the full measure dm_{full}, built from $|\zeta(\frac{1}{2}+is)|^2$, and the on-critical measure $dm_{\text{full,on}}$, built from the Hadamard product with every zero forced onto the critical line. We study their difference, the signed measure $d\nu = dm_{\text{full}} - dm_{\text{full,on}}$, and prove that it is the exact obstruction to the Riemann Hypothesis. Our central unconditional result is a *dominance theorem*: the density $R(s) = |\zeta(\frac{1}{2}+is)/\zeta_{\text{on}}(\frac{1}{2}+is)|^2$ satisfies $R \geq 1$ for all real s, with equality if and only if every nontrivial zero lies on the line; hence $d\nu \geq 0$ unconditionally, and $d\nu = 0 \iff$ RH. From this single non-negative measure we derive a synthesis: eight a priori different conditions on ζ — a zero measure, the vanishing of a Nevanlinna functional, a Jensen excess, a relative entropy, a conditional-positive trace, and the others — are pairwise equivalent, all reducing to $d\nu = 0$. We give the quantitative lower bound that an off-critical zero forces on the trace $T_\lambda = \int W_\lambda \, d\nu$ (second order in the off-line displacement), and we are careful about the one direction that is not unconditional: recovering $d\nu$ from the trace family $\{T_\lambda\}$ requires the totality of the Abel kernels in $L^2(d\nu)$, which we state as an explicit completeness hypothesis and connect to the moment-rigidity criterion for RH proved elsewhere; we do not use, and we explicitly avoid, any pointwise positivity of the Abel kernel as a quadratic form. We frame the obstruction through the function-field analogue, where the same measure vanishes by Weil's three structural pillars (a polarized cohomology, a Frobenius, a Lefschetz pairing), and isolate the missing pillar for ζ as the Hadamard barrier: the Hadamard product delivers $d\nu \geq 0$ but not the completeness or the positivity that would force $d\nu = 0$. Nothing here assumes or proves the Riemann Hypothesis; the contribution is a clean non-negative invariant whose vanishing is RH, together with a precise account of what its vanishing would require.

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1. Introduction

The zeta spectral triple represents ζ through a self-adjoint Jacobi operator whose spectral data encode the nontrivial zeros. Two measures arise naturally. The *full* spectral measure dm_{full} carries the actual modulus $|\zeta(\frac{1}{2}+is)|^2$; the *on-critical* measure $dm_{\text{full,on}}$ carries the modulus of the function ζ_{on} obtained from the Hadamard product of ζ by relocating every zero onto the critical line. If the Riemann Hypothesis (RH) holds, $\zeta = \zeta_{\text{on}}$ and the two coincide; in general they differ, and the difference

$$d\nu = dm_{\text{full}} - dm_{\text{full,on}}$$

is a signed measure on $\mathbb{R}$ that vanishes exactly when RH holds. This paper is a study of $d\nu$ as an invariant in its own right.

The main unconditional fact is that $d\nu$ has a sign. Writing $R(s) = |\zeta(\frac{1}{2}+is)/\zeta_{\text{on}}(\frac{1}{2}+is)|^2$ for the density of dm_{full} relative to $dm_{\text{full,on}}$, we prove $R(s) \geq 1$ for all real s (Theorem 3.1), with equality iff there are no off-critical zeros; thus $d\nu \geq 0$ unconditionally, and $d\nu = 0 \iff$ RH (Corollary 3.2). The mechanism is elementary and exact: each off-critical zero, together with the three partners forced on it by the functional equation and conjugation, contributes to R a factor ≥ 1 that degenerates to 1 precisely when the off-line displacement vanishes.

This non-negative measure organizes a number of previously separate criteria. We prove a synthesis theorem (Theorem 4.1): eight conditions — among them the vanishing of a Nevanlinna transform of $d\nu$, the vanishing of a Jensen excess on a disk model of ζ, the vanishing of a relative entropy, and the vanishing of the conditional-positive trace family $\{T_\lambda\}$ — are all equivalent to $d\nu = 0$, hence to RH. We then turn to the trace $T_\lambda = \int W_\lambda \, d\nu$ with W_λ the positive Abel kernel. Because $d\nu \geq 0$ and $W_\lambda > 0$, each $T_\lambda \geq 0$ *without cancellation*, and a single off-critical zero forces a strictly positive lower bound that is second order in its displacement (Proposition 5.1). The one delicate point — which we treat with care — is the reverse implication: that $T_\lambda = 0$ for all λ forces $d\nu = 0$. This requires the totality of $\{W_\lambda\}$ in $L^2(d\nu)$; we isolate it as an explicit completeness hypothesis and relate it to the moment-rigidity criterion for RH established through orthogonal polynomials on the line [1]. We do *not* appeal to any pointwise sign of W_λ as a quadratic form; the only positivity used is that of $d\nu$ itself and of the scalar test functions.

Finally we locate the obstruction structurally. In the function-field analogue the same measure vanishes *for a proven reason*: Weil's three pillars — a polarized cohomology with a positive intersection form, a Frobenius whose eigenvalues are the zeros, and a Lefschetz/Hodge pairing forcing them onto the critical circle. For ζ the first two have candidate analogues but the polarization is absent; what remains is the *Hadamard barrier*: the Hadamard product supplies $d\nu \geq 0$ but supplies neither the completeness nor the positive intersection form that would force $d\nu = 0$. We make this comparison precise (§6–§7). Nothing in the paper assumes or proves RH.

2. The two measures and the density R

Definition 2.1 (Full and on-critical measures)**.** Let dm_∞ be the reference (archimedean) measure of the spectral triple. The full and on-critical spectral measures are the absolutely continuous measures

$$dm_{\text{full}} = |\zeta(\tfrac{1}{2}+is)|^2 \, dm_\infty, \qquad dm_{\text{full,on}} = |\zeta_{\text{on}}(\tfrac{1}{2}+is)|^2 \, dm_\infty,$$

where ζ_{on} is defined by the Hadamard product of ζ with every nontrivial zero $\rho = \frac{1}{2}+\delta+i\gamma$ replaced by its on-critical projection $\frac{1}{2}+i\gamma$. The density of dm_{full} relative to $dm_{\text{full,on}}$ is $R(s) = |\zeta(\frac{1}{2}+is)/\zeta_{\text{on}}(\frac{1}{2}+is)|^2$, and

$$d\nu = dm_{\text{full}} - dm_{\text{full,on}} = (R-1)\, dm_{\text{full,on}}.$$

3. The dominance theorem

Theorem 3.1 (Unconditional dominance of ζ over ζ_{on})**.** *For all $s \in \mathbb{R}$, $R(s) \geq 1$, with equality for all s if and only if* RH *holds.*

Proof. The nontrivial zeros are symmetric under conjugation and $s \mapsto 1-s$, so each off-critical zero $\rho_0 = \frac{1}{2} + \delta_0 + i\gamma_0$ ($\delta_0 \neq 0$) belongs to a quadruple $\{\rho_0, \bar{\rho}_0, 1-\rho_0, 1-\bar{\rho}_0\}$; the on-critical replacement moves all four to $\frac{1}{2} \pm i\gamma_0$ (each with multiplicity two). On the critical line $s = \frac{1}{2} + it$, the ratio ζ/ζ_{on} is the product over off-critical quadruples of

$$\frac{(\frac{1}{2}+it-\rho_0)(\frac{1}{2}+it-\bar{\rho}_0)(\frac{1}{2}+it-(1-\rho_0))(\frac{1}{2}+it-(1-\bar{\rho}_0))}{(\frac{1}{2}+it-(\frac{1}{2}+i\gamma_0))^2(\frac{1}{2}+it-(\frac{1}{2}-i\gamma_0))^2}.$$

Computing moduli, $|\frac{1}{2}+it-\rho_0|^2 = \delta_0^2 + (t-\gamma_0)^2$, and likewise the four numerator factors give $\delta_0^2 + (t \mp \gamma_0)^2$ (twice each), while the denominators give $(t-\gamma_0)^2$ and $(t+\gamma_0)^2$ (squared). Hence the squared modulus of the single-quadruple contribution is

$$\frac{(\delta_0^2 + (t-\gamma_0)^2)^2(\delta_0^2 + (t+\gamma_0)^2)^2}{(t-\gamma_0)^4(t+\gamma_0)^4} \geq 1,$$

since $\delta_0^2 + u^2 \geq u^2$ for every u, with equality iff $\delta_0 = 0$. The product over quadruples converges absolutely (the off-critical zeros, if any, obey the global density bound $N(T) = O(T\log T)$, and all factors are ≥ 1), giving $R(s) \geq 1$, with equality for all s iff every $\delta_0 = 0$, i.e. RH. $\square$

Corollary 3.2 ($d\nu$ is a non-negative invariant for RH)**.** *Unconditionally $d\nu = (R-1)\,dm_{\text{full,on}} \geq 0$. Moreover $d\nu = 0 \iff$* RH*, and $d\nu$ is even ($d\nu(s) = d\nu(-s)$, from $\xi(s) = \xi(1-s)$) and finite, with*

$$\int_{\mathbb{R}} d\nu = \sum_j \frac{4\delta_j^2}{\gamma_j^2 + \delta_j^2} + O\Big(\sum_j \frac{\delta_j^4}{\gamma_j^4}\Big) < \infty,$$

the sum over off-critical zeros, convergent because $\sum_j \delta_j^2/\gamma_j^2 < \infty$ by the zero-counting estimate.

Proof. Non-negativity and the equivalence are Theorem 3.1. Parity is the functional equation. For finiteness, each quadruple contributes $(R-1)$ of order δ_j^2/γ_j^2, summable by $N(T) = O(T\log T)$ and $\sum_j \delta_j^2/\gamma_j^2 < \infty$. $\square$

Remark 3.3 (The curve case). For a smooth projective curve over a finite field, the analogous measure vanishes identically: every zero of the zeta function already lies on the critical circle, so the product analogous to R has no nontrivial factors. There $d\nu = 0$ is *known* from Weil's proof; for ζ we know only $d\nu \geq 0$. The whole difficulty is the gap between these two statements, made precise in §6–§7.

4. The synthesis: eight equivalent criteria

Several criteria that arise independently — from a Nevanlinna transform, a Jensen excess on a disk model, a relative entropy, and the spectral trace — all reduce to $d\nu = 0$.

Theorem 4.1 (Eight equivalent criteria)**.** *The following are pairwise equivalent.*

(1) (RH) *every nontrivial zero of ζ has* $\operatorname{Re} s = \frac{1}{2}$*;*
(2) (Zero measure) $d\nu = 0$*;*
(3) (Nevanlinna) $\mathcal{G}(z) = \int_{\mathbb{R}} \frac{s+z}{s-z}\,d\nu(s)$ *vanishes identically on* $\{\operatorname{Im} z > 0\}$*;*
(4) (Trace) $T_\lambda = \int_{\mathbb{R}} W_\lambda\,d\nu = 0$ *for every $\lambda > 0$, where $W_\lambda > 0$ is the Abel kernel;*
(5) (Jensen excess) $\delta = \sum_{\operatorname{Re}\rho > 1/2} \log(|\rho|/|\rho-1|) = 0$*;*
(6) (Variational) *the squared distance $\widetilde{\delta}_\lambda^2$ from $|\zeta|^2 dm_\infty$ to the on-critical variety vanishes for every λ;*
(7) (Relative entropy) $\mathrm{KL}(dm_{\text{full}} \,\|\, dm_{\text{full,on}}) = 0$*;*

(8) (CCM distance) $d_{\mathrm{CCM}} = \sup_{\lambda>0} T_\lambda = 0$.

The equivalences (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) $\Leftrightarrow$ (5) $\Leftrightarrow$ (7) *are unconditional. The equivalences involving the trace,* (2) $\Leftrightarrow$ (4) $\Leftrightarrow$ (6) $\Leftrightarrow$ (8)*, hold under the completeness hypothesis of §5* (*the totality of* $\{W_\lambda\}$ *in* $L^2(d\nu)$)*; the forward directions* (2) $\Rightarrow$ (4), (6), (8) *are unconditional.*

Proof. (1) $\Leftrightarrow$ (2) is Corollary 3.2. (2) $\Leftrightarrow$ (3): the Nevanlinna kernels $\{(s+z)/(s-z) : \operatorname{Im} z > 0\}$ are total in $L^2(d\nu)$, so $\mathcal{G} \equiv 0 \iff d\nu = 0$. (1) $\Leftrightarrow$ (5): by Jensen's formula on the disk model $F(z) = \xi(z/(z-1))$, the excess $\delta \geq 0$ with equality iff F is zero-free in $|z| < 1$, iff RH. (2) $\Leftrightarrow$ (7): $\mathrm{KL}(dm_{\mathrm{full}} \| dm_{\mathrm{full,on}}) = \int \log R \, dm_{\mathrm{full,on}}$, and $\log x \geq (x-1) - \frac{1}{2}(x-1)^2$ with $R \geq 1$ gives $\mathrm{KL} = 0 \iff R = 1$ a.e. $\iff d\nu = 0$. For the trace equivalences: $d\nu \geq 0$ and $W_\lambda > 0$ give $T_\lambda \geq 0$, so $d\nu = 0 \Rightarrow T_\lambda = 0 \ \forall\lambda$ (forward, unconditional); the converse $T_\lambda = 0 \ \forall\lambda \Rightarrow d\nu = 0$ is exactly the completeness hypothesis of §5. (2) $\Leftrightarrow$ (6) is the same statement, since $\widetilde{\delta}_\lambda^2 = T_\lambda$ up to a positive constant, and (4) $\Leftrightarrow$ (8) because $d_{\mathrm{CCM}} = \sup_\lambda T_\lambda$ vanishes iff every T_λ does. □

5. The trace, its lower bound, and the completeness hypothesis

Proposition 5.1 (Lower bound under ¬RH)**.** *Suppose* ¬RH*, with off-critical zeros* $\rho_j = \frac{1}{2} + \delta_j + i\gamma_j$*. For each* $\lambda > 0$ *and each* j *with* $|\gamma_j| \leq \lambda$*,*

$$T_\lambda \geq 2 \sum_{|\gamma_j| \leq \lambda} \delta_j^2 \, K_\lambda(\gamma_j), \qquad K_\lambda(\gamma_j) = W_\lambda(\gamma_j)\, dm_{\mathrm{full,on}}(\{\gamma_j\}) > 0$$

for λ *large enough; hence the right-hand side is strictly positive. The trace detects an off-critical zero to second order in its displacement* δ_j*.*

Proof. By Theorem 3.1, near γ_j the density satisfies $R(t) - 1 \geq (\delta_j^2 + (t-\gamma_j)^2)^2/(t-\gamma_j)^4 - 1 \geq 2\delta_j^2/(t-\gamma_j)^2 + O(\delta_j^4)$; integrating $W_\lambda\,(R-1)\,dm_{\mathrm{full,on}}$ over a unit window at γ_j and summing the (non-negative) contributions gives the bound. □

The decisive point is the reverse implication. Since $d\nu \geq 0$ and $W_\lambda > 0$, the trace is a *linear* functional of a non-negative measure tested against a positive kernel; there is no sign cancellation, and the only question is whether the kernels $\{W_\lambda\}$ *see all of* $d\nu$.

Definition 5.2 (Completeness hypothesis)**. (C)** The Abel kernels $\{W_\lambda : \lambda > 0\}$ are total in $L^2(d\nu)$ for every non-negative even finite measure $d\nu$ of the form $(R-1)dm_{\mathrm{full,on}}$.

Theorem 5.3 (Trace criterion under (C))**.** *Under hypothesis* (C), $T_\lambda = 0$ *for all* $\lambda > 0$ *if and only if* RH*. Unconditionally,* RH $\Rightarrow T_\lambda = 0 \ \forall\lambda$*, and a single off-critical zero forces some* $T_\lambda > 0$ (*Proposition 5.1*)*.*

Proof. RH $\Rightarrow d\nu = 0 \Rightarrow T_\lambda = 0$ for all λ (Corollary 3.2). Under (C), $T_\lambda = 0 \ \forall\lambda$ means $d\nu \perp W_\lambda$ for all λ, hence $d\nu = 0$ by totality, hence RH. □

Remark 5.4 (What is and is not used). We emphasize that Theorem 5.3 uses *only* $d\nu \geq 0$ (the dominance theorem) and the scalar positivity $W_\lambda > 0$. It does *not* use, and we do not assert, any positivity of W_λ as a quadratic form (a kernel positivity); such a pointwise-kernel claim is false in general and plays no role here. Hypothesis (C) is a genuine analytic input — a completeness of the Abel system in $L^2(d\nu)$ — and is the honest residue of the trace criterion. It is equivalent, via the orthogonal-polynomial realization of $d\nu$, to the moment-rigidity criterion proved in [1], where RH is shown equivalent to the constancy of a sequence of Christoffel-type functionals; (C) is the statement that the trace family recovers those functionals.

6. The function-field mirror: Weil's three pillars

The reason $d\nu$ vanishes in the function-field case is structural and instructive.

Theorem 6.1 (Function-field trace vanishes)**.** *For the zeta function Z_C of a smooth projective curve C over a finite field, the analogue $d\nu^C$ of $d\nu$ is identically zero.*

The vanishing rests on three pillars, of which ζ has analogues of the first two but not the third:

(P1) a finite-dimensional cohomology $H^1(C)$ carrying the zeros as Frobenius eigenvalues (*for* ζ: a candidate spectral realization exists — the Jacobi operator of the spectral triple — but it is infinite-dimensional and its self-adjointness is the analogue of, not a substitute for, the Frobenius);

(P2) a Lefschetz/explicit-formula trace relating the eigenvalues to point counts (*for* ζ: the Riemann–Weil explicit formula plays this role);

(P3) a **polarization**: a positive intersection form on $H^1(C)$ (the Hodge index / Riemann form) that forces the Frobenius eigenvalues onto the critical circle (*for* ζ: absent — this is exactly the missing positivity).

Remark 6.2. The measure $d\nu$ is the analytic shadow of the failure of: in the curve case the polarization forces $R \equiv 1$ (all zeros on the circle, $d\nu^C = 0$); for ζ the Hadamard product gives only $R \geq 1$, the one-sided remnant of a polarization that is not known to be two-sided. The trace criterion (Theorem 5.3) and the moment rigidity of [1] are two ways of asking for the missing pillar.

7. The Hadamard barrier

Definition 7.1 (Hadamard barrier)**.** The *Hadamard barrier* is the gap between the two facts: (a) the Hadamard product of ξ yields, unconditionally, the one-sided bound $R \geq 1$, i.e. $d\nu \geq 0$; (b) forcing $d\nu = 0$ requires either the completeness (C) of the Abel system in $L^2(d\nu)$ or, equivalently, a positive intersection form on the spectral realization. The Hadamard product delivers (a) but not (b).

Proposition 7.2 (The barrier is genuine)**.** *The implication $d\nu \geq 0 \Rightarrow d\nu = 0$ is false for general non-negative even finite measures; it holds for the specific $d\nu = (R-1)dm_{\text{full,on}}$ if and only if* RH*. Hence no purely soft argument from $d\nu \geq 0$ can close the gap: the additional input is exactly of the strength of* RH*, and is supplied by neither the Hadamard product nor the explicit formula alone.*

Proof. A non-negative even finite measure need not vanish, so $d\nu \geq 0$ alone is insufficient. By Corollary 3.2, $d\nu = (R-1)dm_{\text{full,on}} = 0 \iff$ RH. Thus any argument deducing $d\nu = 0$ supplies RH, so it cannot be soft; in particular it is not a consequence of the structural facts (dominance, explicit formula, self-adjointness of the Jacobi operator) that hold unconditionally. □

8. Conclusion

The difference of the two spectral measures of the zeta spectral triple is a single, clean, non-negative invariant: $d\nu \geq 0$ unconditionally (Theorem 3.1), with $d\nu = 0$ exactly the Riemann Hypothesis (Corollary 3.2). It unifies a family of criteria (Theorem 4.1) and is detected to second order by a positive trace family (Proposition 5.1). The criterion's hard direction is honestly isolated: it is the completeness (C) of the Abel kernels in $L^2(d\nu)$ (Theorem 5.3, Remark 5.4), equivalent to the moment-rigidity criterion of [1], and we use no pointwise kernel positivity. The function-field mirror (§6) names the missing ingredient as Weil's polarization pillar, and the Hadamard barrier (§7) proves that no soft argument from $d\nu \geq 0$ can supply it: the input needed is exactly of the strength of RH. The paper neither assumes nor proves the Riemann Hypothesis; it provides a sign-definite invariant whose vanishing is RH and a precise map of what its vanishing demands.

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